

MONTE CARLO ASSIGNMENT 1

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1. RANDOM NUMBER GENERATION

Let X be a random variable on $\mathbb{R}$ with density f_X and invertible distribution function F_X . Here f_X , F_X , and the inverse F_X^{-1} are assumed to be known. Let $I = (a, b)$ be an interval such that $\mathbb{P}(X \in I) > 0$.

(a) By the definition of conditional probability

$$F_{X|X \in I}(x) = P(X \leq x | X \in I) = \frac{P(X \leq x, X \in I)}{P(X \in I)}.$$

Since $X \in I = (a, b)$, we can rewrite the conditional CDF as

$$F_{X|X \in I}(x) = \frac{F_X(x) - F_X(a)}{F_X(b) - F_X(a)}, \quad \text{for } a \leq x \leq b.$$

By differentiating the conditional CDF which gives the conditional density:

$$f_{X|X \in I}(x) = \frac{d}{dx} F_{X|X \in I}(x) = \frac{f_X(x)}{F_X(b) - F_X(a)}, \quad a \leq x \leq b.$$

(b) Assume that $F_{X|X \in I}(x) = u = \frac{F_X(x) - F_X(a)}{F_X(b) - F_X(a)}$. We aim to find the inverse $F_{X|X \in I}^{-1}(u)$. By rearranging in x we get

$$x = F_{X|X \in I}^{-1}(u) = F_X^{-1}(F_X(a) + u(F_X(b) - F_X(a))).$$

The random variable X can be simulated using the inversion sampling method. First draw $U \sim \mathcal{U}(0, 1)$, where $\mathcal{U}(0, 1)$ is the uniform distribution over $(0, 1)$. Then compute

$$X = F_X^{-1}(F_X(a) + u(F_X(b) - F_X(a))).$$

In this way we can sample X such that it always lies within (a, b) while preserving its original distribution.

2. POWER PRODUCTION OF A WIND TURBINE

Given a proposal to build a Vestas V164 wind turbine in a specific location in the southern part of Öresund, we aim to investigate potential power outputs of said wind turbine. The wind speeds at the building site may be modelled by a Weibull distribution, with parameters varying over each month. The Monte Carlo method will be used to approximate 95% confidence intervals for the expected amount of power (given in W in section 2

and MW in section 3) output for each month of the year. The confidence intervals will be in the following form

$$\left(\tau_N - \lambda_{\alpha/2} \frac{\sigma(\phi)}{\sqrt{N}}, \tau_N + \lambda_{\alpha/2} \frac{\sigma(\phi)}{\sqrt{N}} \right),$$

where τ_N is an estimation of the expected power output, $\lambda_{\alpha/2}$ is the 97.5th percentile point of the standard normal distribution (with an approximate numerical value of 1.96), $\sigma(\phi)$ is the standard deviation of the objective function ϕ (which in this case is the power output of the turbine), and finally $\sqrt{N}$ is the square of the sample size for each simulation. In this report, N is chosen to be 10000. Thus, to create the approximate confidence intervals 95%, τ_N and $\sigma(\phi)$ are needed.

First, the standard Monte Carlo method will be used. From the Law of Large Numbers, it is known that

$$\frac{1}{N} \sum_{i=1}^N \phi(X_i) \longrightarrow \mathbb{E}(\phi(X)) \quad \text{as } N \rightarrow \infty \quad (a.s.).$$

Having chosen N to be a large number (10000), τ_N is then given by

$$\tau_N = \frac{1}{N} \sum_{i=1}^N \phi(x_i),$$

where each x_i is drawn independently from a Weibull distribution with parameters chosen according to the given month. Further, the standard deviation of the objective function is given by the square root of MATLAB inbuilt function "std", which is the usual bias-corrected estimator for the variance (as given in the 2nd lecture notes). The obtained 95% approximate confidence intervals, and their widths, for each month of the year are as follows.

Table 1: 95% approximate confidence intervals		
Month	Interval	Interval width
January	(4645045.4754, 4690532.8782)	45487.4027
February	(4118952.6605, 4163103.3409)	44150.6804
March	(3815758.4657, 3858715.6072)	42957.1415
April	(2977787.0779, 3017406.8581)	39619.7802
May	(2848932.0512, 2887795.3772)	38863.326
June	(3062275.9907, 3102459.1932)	40183.2026
July	(2841699.0175, 2880567.9894)	38868.9719
August	(3072611.4294, 3112827.3893)	40215.9599
September	(3716883.7884, 3759632.917)	42749.1287
October	(4202943.0601, 4247971.4515)	45028.3914
November	(4646983.1666, 4692435.0822)	45451.9156
December	(4655270.7719, 4700713.7961)	45443.0242

Table of 95% confidence intervals obtained from standard Monte Carlo simulation.

Now, a truncated Weibull distribution function will be used in order to get more narrow confidence intervals. In this case the domain is restricted to $[3.5, 25]$, as the turbine only produces power with wind speeds within this interval. Results from section 1 will now be utilized. Before, each random number was independently drawn using from the Weibull distribution using MATLAB's inbuilt function *wblrnd*, while now each number is picked according to

$$x = F_X^{-1}(F_X(a) + u(F_X(b) - F_X(a))),$$

where u is a number drawn from the uniform distribution on the interval $(0, 1)$, and $a = 3.5$, $b = 25$. (The derivation of this formula was given in section 1). The procedure now follows as before when the standard Monte Carlo method was used. Following are the 95% approximated confidence intervals using a truncated Weibull distribution.

Table 2: Truncated 95% approximate confidence intervals		
Month	Interval	Interval width
January	(4619328.4543, 4657938.8796)	38610.4253
February	(4115596.3231, 4152822.7006)	37226.3775
March	(3796471.5102, 3832489.7163)	36018.2061
April	(2990401.4965, 3022338.7657)	31937.2692
May	(2852948.6024, 2883970.9621)	31022.3598
June	(3064930.5351, 3097353.1545)	32422.6194
July	(2868554.3601, 2899715.7424)	31161.3824
August	(3067209.7224, 3099631.5351)	32421.8126
September	(3743689.0919, 3779455.2839)	35766.192
October	(4211363.4999, 4248847.3209)	37483.8211
November	(4635898.7632, 4674499.4421)	38600.679
December	(4631420.4396, 4670035.5127)	38615.0731

Table of truncated 95% confidence intervals obtained from Standard Monte Carlo simulation.

Having restricted the distribution function to be between 3.5 and 25, the widths of the truncated confidence intervals are smaller.

- (b) A control variate will now be introduced to decrease the variance (and thus the standard deviation) to further narrow the confidence intervals for the estimated power output of the wind turbine, for each month of the year. The variance reduction works as follows.

Given a random variable Y for which the expectation $\mathbb{E}(Y)$ is known and for which simulation at the same level as for $\phi(X)$ is possible, it can be used as a control variate. Set

$$Z = \phi(X) + \beta(Y - \mathbb{E}(Y)),$$

such that

$$\mathbb{E}(Z) = \mathbb{E}(\phi(X) + \beta(Y - \mathbb{E}(Y))) = \mathbb{E}(\phi(X)).$$

Then, taking the variance of Z and minimizing with respect to β yields that

$$\beta = -\frac{\mathbb{C}(\phi(X), Y)}{\mathbb{V}(Y)}$$

is the optimal coefficient to be used in Z for variance reduction. The control variate which will be used in the setting of this report is the stochastic variable V which documents wind speeds. Thus, the V 's in the above formulas are replaced with the variable Y , and $\phi(X)$ is the function of power output. The expectation of V is given by

$$\mathbb{E}(V^m) = \Gamma(1 + m/k)\lambda^m, \quad m > -k,$$

where the constants k and λ are adjusted according to the given Weibull distribution of each month. Having found the expected value $\mathbb{E}(V)$ of V and the optimal constant β , the following variance reduced 95% normal approximate confidence intervals are obtained. As a consequence of the lowered variance, the widths of the confidence intervals have been further reduced.

Table 3: 95% approximate confidence intervals using a control variate		
Month	Interval	Interval width
January	(4648456.3107, 4668955.3344)	20499.0237
February	(4132755.7954, 4149043.8935)	16288.0981
March	(3817781.3372, 3832097.1349)	14315.7977
April	(2998225.7951, 3010581.5306)	12355.7355
May	(2864483.3752, 2876353.9321)	11870.5569
June	(3070613.5673, 3083213.5617)	12599.9944
July	(2865447.2229, 2877455.1203)	12007.8974
August	(3074567.1807, 3086897.9728)	12330.7921
September	(3757652.8209, 3771788.2795)	14135.4587
October	(4218468.9081, 4237289.2012)	18820.2931
November	(4648318.9565, 4668978.4071)	20659.4506
December	(4648681.0399, 4669118.4862)	20437.4463

Table of variance reduced 95% confidence intervals obtained from Monte Carlo simulation with a control variate.

- (c) Importance sampling will now be used to create new 95% approximate confidence intervals for the power outputs for the months of the year. To do this, an instrumental density function g must be chosen such that the already given density function (in this case from a Weibull distribution) is zero whenever the instrumental density g is zero. The expected value of the objective function may then be rewritten as

$$\begin{aligned} \mathbb{E}_f(\varphi(X)) &= \int_X \varphi(x)f(x) dx \\ &= \int_{g(x)>0} \varphi(x) \frac{f(x)}{g(x)} g(x) dx \\ &= \mathbb{E}_g(\varphi(X)\omega(X)), \end{aligned}$$

where $\omega(X)$ is a function given by

$$\omega : \{x \in X : g(x) > 0\} \ni x \mapsto \frac{f(x)}{g(x)}.$$

After having chosen the density g , the procedure follows a standard Monte Carlo simulation, as before (this time generating random numbers from the distribution of g). The goal is to choose g such that the function

$\varphi(x) \frac{f(x)}{g(x)} g(x)$ is as close to constant as possible, such that the variance is as close to 0 as possible. To do this, g is chosen to be similar to $\varphi(x)f(x)$. For each month the function $\varphi(x)f(x)$ is similar to a normal distribution with a certain mean and standard deviation.

Thus, for each month, g is chosen as a normal distribution with a mean and variance to match the given $\varphi(x)f(x)$ as closely as possible, where the differences for each month are due to parameter changes in the Weibull density $f(x)$. The mean of g is chosen so that the peak of the chosen distribution aligns with the peak of the $\varphi(x)f(x)$ function. The variance of g is then chosen so that the curves match as well as possible, to minimize variance (by the logic explained above). An example of this is given in Figure 1 below, where the function $\varphi(x)f(x)$ follows the parameters corresponding to January and where g has been chosen as a density of a normal distribution matching $\varphi(x)f(x)$. Specifically, this g is chosen to be the density corresponding to a normal distribution with mean 12, standard deviation 4.5, multiplied by 5080000 to get a matching amplitude. (Note that this g is *not* a distribution, as its amplitude is magnified. Rather, it serves as a figure to show how comparisons can be made.)

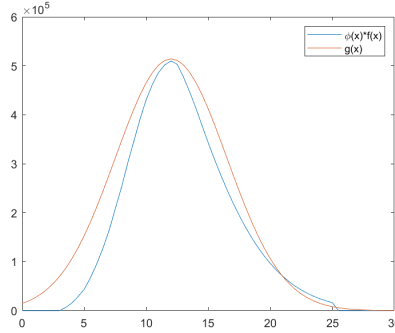


FIGURE 1. Suggested g function (in red), Power Curve $P(v)$ times Weibull Distribution for January $f(v)$ (in blue).

It is visible that the functions are closely matching. However, due to the asymmetry of the $\varphi(x)f(x)$ functions (and the symmetry of the Gaussian density), there will exist some discrepancies, as seen in Figure 1. The same logic and procedure follows for the remaining months. The following table contains confidence intervals obtained from the Monte Carlo simulation using importance sampling, where the instrumental density has been manually chosen as a specific Normal density for each month, to match the pertaining $\varphi(x)f(x)$ function.

Table 4: 95% approximate confidence intervals using importance sampling		
Month	Interval	Interval width
January	(4648702.0769, 4665430.0194)	16727.9425
February	(4141143.6977, 4155005.9306)	13862.2329
March	(3817207.6792, 3831102.523)	13894.8437
April	(3002255.0056, 3012027.4976)	9772.4921
May	(2859858.0977, 2868180.8123)	8322.7146
June	(3074128.3454, 3084445.3708)	10317.0254
July	(2864160.7153, 2872476.1592)	8315.4439
August	(3079473.865, 3089707.8911)	10234.0261
September	(3754550.4773, 3767146.313)	12595.8358
October	(4212856.4143, 4226257.0986)	13400.6843
November	(4663976.7436, 4680882.4467)	16905.7031
December	(4655597.0745, 4672391.0783)	16794.0038

Table of 95% confidence intervals obtained from Monte Carlo simulation using importance sampling.

The confidence intervals are much narrower using importance sampling, compared to the previous variance reduction methods. Some months having a slightly better estimate is due to the pertaining $\varphi(x)f(x)$ function being more symmetric. Since the instrumental densities are Gaussian, and thus symmetric, the estimates for these months are better.

- (d) Another technique will now be used for variance reduction – antithetic sampling. In line with Lecture notes 4, let $V = \phi(X)$ such that $\tau = \mathbb{E}(\phi(X)) = \mathbb{E}(V)$ and assume that we can generate a variable $\tilde{V}$ such that $\mathbb{E}(\tilde{V}) = \tau$, $\mathbb{V}(\tilde{V}) = \mathbb{V}(V)$ and that $\tilde{V}$ can be simulated at the same complexity as V . Then introduce

$$W = \frac{V + \tilde{V}}{2}.$$

Then

$$\mathbb{E}(W) = \tau$$

and

$$\mathbb{V}(W) = \frac{1}{2(\mathbb{V}(V) + \mathbb{C}(V, \tilde{V}))}.$$

To get a reduced variance, $\tilde{V}$ needs to be chosen such that $\tilde{V}$ and V are negatively correlated. This is done using the theorem pertaining to antithetic sampling given in Lecture notes 4. In practice, let F be a distribution function and ϕ a monotone function. Then, for $U \sim \mathcal{U}(0, 1)$, $T(u) = 1 - u$. Then $\varphi(u) = \phi(F^{-1}(u))$ yields

$$V = \phi(F^{-1}(U)) \stackrel{d}{=} \tilde{V} = \phi(F^{-1}(1 - U))$$

and

$$\mathbb{C}(V, \tilde{V}) \leq 0.$$

Thus, the expected power output can be approximated with a lower variance after acquiring V and $\tilde{V}$. The results of simulation using antithetic

sampling are given below, in table 5.

Month	Interval	Interval width
January	(4659111.5382, 4665047.4064)	5935.8682
February	(4139056.306, 4146986.0998)	7929.7938
Mars	(3817499.0963, 3827292.3446)	9793.2483
April	(3009028.0917, 3023052.0178)	14023.9261
May	(2867341.5695, 2881724.9893)	14383.4198
June	(3070990.9835, 3084783.6704)	13792.6869
July	(2867224.8487, 2881623.0437)	14398.1949
August	(3069870.0639, 3083625.5778)	13755.5138
September	(3757863.7602, 3768021.8047)	10158.0446
October	(4221883.942, 4229992.231)	8108.2891
November	(4659952.287, 4665846.3113)	5894.0243
December	(4656537.4702, 4662581.8513)	6044.3811

Table of variance reduced 95% confidence intervals obtained from Monte Carlo simulation with antithetic sampling.

It is clear that the results from Monte Carlo simulation using antithetic sampling has led to the best results in variance reduction, comparing with the previous variance reduction methods. This is seen by the drastically reduced widths of the confidence intervals in Table 5 above.

- (e) A table containing the probabilities of the wind turbine producing power given the months of the year will now be given. For each month, a random sample of size $N = 10000$ is drawn from a Weibull distribution with parameters corresponding to each given month. The power function (the "objective function") is then applied to the sample for each month. The probability of power is then given by the ratio of the number of non-zero powers and the sample size $N = 10000$. Table 6 with the probability of the turbine producing power for each month is given below.

Month	Probability of power
January	0.8931
February	0.87809
March	0.86472
April	0.81305
May	0.80647
June	0.8171
July	0.80489
August	0.81587
September	0.8615
October	0.86671
November	0.89287
December	0.89433

- (f) The average ratio of actual wind turbine output to total wind power will now be given, again in terms of 95% confidence intervals. The ratio that is

to be estimated is

$$\frac{\mathbb{E}P(V)}{\mathbb{E}P_{tot}(V)}.$$

This is done by calculating the mean of the power function (objective function) $P(V)$, where V is generated from the Weibull distribution with parameters corresponding to each month, and $P_{tot}(V)$ is given by

$$P_{tot}(v) = \frac{1}{2}\rho\pi\left(\frac{d^2}{4}\right)v^3$$

, where $d = 164$, $\rho = 1.225\text{kg}/\text{m}^3$, and $v^3 = \Gamma(1 + \frac{3}{k})\lambda^3$. The estimated 95% confidence intervals are given in Table 6 below.

Table 6: 95% approximate confidence intervals for power ratios		
Month	Interval	Interval width
January	(0.22629, 0.22851)	0.0022186
February	(0.26195, 0.26477)	0.0028155
March	(0.28362, 0.28683)	0.0032069
April	(0.32129, 0.32556)	0.0042709
May	(0.32963, 0.33412)	0.0044915
June	(0.31439, 0.31853)	0.0041396
July	(0.32947, 0.33397)	0.0044965
August	(0.31479, 0.31893)	0.0041443
September	(0.28839, 0.29169)	0.0032977
October	(0.23793, 0.24048)	0.0025488
November	(0.22623, 0.22845)	0.0022219
December	(0.2262, 0.22843)	0.0022201

Table of 95% confidence intervals for the Power Ratio and Interval Width for Each Month.

- (g) The final step in this section consists of calculating the *capacity factor* and the *availability factor*. The capacity factor is the ratio of power output to the maximum possible power output during a certain period, and the availability factor during a certain period is the amount of time that electricity is produced during the period divided by the length of the period. Thus, the capacity factor is given by

$$\frac{\mathbb{E}P(V)}{9.5 \times 10^6}$$

and the availability factor is given by the mean of the ratios given in section 2.e). The capacity factor and availability factor are given in Table 7 below.

Table 7: Capacity factor and Availability factor	
Capacity factor	Availability factor
0.39373	0.85016

3. COMBINED POWER PRODUCTION OF TWO WIND TURBINES

- (a) We need to estimate the expected combined power output of two wind turbines: $\mathbb{E}[P(V_1) + P(V_2)]$, where V_1 and V_2 follow a bivariate Weibull distribution. Since both turbines have the same marginal distribution and same expectation, the problem reduces to one dimensional.

$$\mathbb{E}[P(V_1) + P(V_2)] = \mathbb{E}[P(V_1)] + \mathbb{E}[P(V_2)] = 2\mathbb{E}[P(V)]$$

By plotting the functions $P(v)$ and $f(v)$ we see that importance sampling is necessary because the two curves are not similar. Thus, our main goal is using importance sampling to estimate:

$$\mathbb{E}[P(V)] = \int_{3.5}^{25} P(v) f_{\text{cond}}(v) dv = \int_{3.5}^{25} P(v) \frac{f_{\text{cond}}(v)}{g(v)} g(v) dv.$$

where $g(v)$ is the instrumental density and $f_{\text{cond}}(v)$ is the conditional Weibull distribution given by: $f_{\text{cond}}(v) = \frac{f(v)}{F(25) - F(3.5)}$ with F the Weibull CDF. We first scale the g function for better observation. From the observation of graph of the weighted function $f(v)P(v)$ in Figure 2, it is similar to a normal distribution, therefore we choose a normal distribution with $\mu_g = 11m/s$ and $\sigma_g = 4.4m/s$, since the peak of the weighted function is around 11.

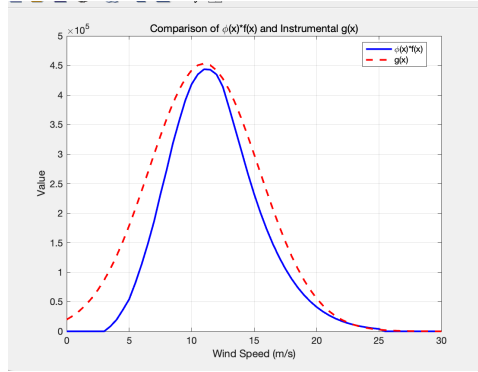


FIGURE 2. Suggested g function and weighted function $P(v)f(v)$.

To make sure the implementation works, we draw 10000 samples from g we can estimate $2\mathbb{E}[P(V)]$ by:

$$2\mathbb{E}[P(V)] \approx \frac{2}{N} \sum_{i=1}^N P(V_{\text{IS},i}) \frac{f_{\text{cond}}(V_{\text{IS},i})}{g(V_{\text{IS},i})},$$

where $V_{\text{IS},i}$ are N samples drawn from the truncated normal instrumental distribution $g(v)$, $\frac{f_{\text{cond}}(V_{\text{IS},i})}{g(V_{\text{IS},i})}$ is the importance weight function and $P(V_{\text{IS},i})$ is the power output function evaluated at each sampled wind speed. Our final estimation is $2E[P(V)] = 7532859.6129W$

- (b) The covariance between $P(V_1)$ and $P(V_2)$ is given by:

$$\mathbb{C}(P(V_1), P(V_2)) = \mathbb{E}[P(V_1)P(V_2)] - (\mathbb{E}[P(V)])^2,$$

since the two wind turbines have the same marginal distribution and we already computed $E[P(V)]$. Thus we need to estimate the joint expectation $\mathbb{E}[P(V_1)P(V_2)]$. We first construct a grid over the wind speed range and check $\phi(x) \cdot \phi(y) \cdot f_{XY}(x, y)$ to see where it peaks, where $\phi(x)$ is the target density. The computation shows that $\mu_g = 12$ is a reasonable choice. The standard deviation for the bivariate normal distribution g is chosen to be 20 after checking the contour plot shown in 3. Note that we choose the σ slightly larger than 4.4×4.4 which is the standard deviation of the marginal distribution from 3(a) squared.

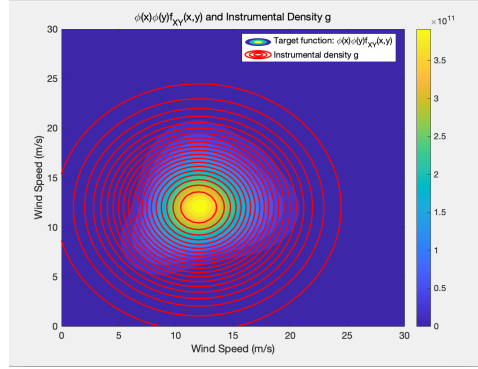


FIGURE 3. Contour graph of weighted function $\phi(x) \cdot \phi(y) \cdot f_{XY}(x, y)$ and instrumental function g .

For more simple observation we assume (x_1, x_2) are having the same value in two dimensions so that we can reduce the problem to one dimensional. Below is the graph showing our suggested instrumental density g and target distribution function.

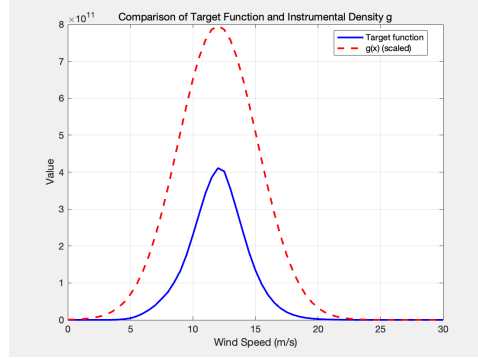


FIGURE 4. Suggested g and target distribution function

Thus we draw 10000 samples from $g(v_1, v_2)$ to estimate:

$$\mathbb{E}[P(V_1)P(V_2)] \approx \frac{1}{N} \sum_{i=1}^N P(V_{1,i})P(V_{2,i}) \frac{f_{cond}(V_{1,i}, V_{2,i})}{g(V_{1,i}, V_{2,i})}$$

Once we obtain $\mathbb{E}[P(V_1)P(V_2)]$ and $\mathbb{E}[P(V)]$ (which is from part (a)), we compute the estimation of covariance is: $6.662 \times 10^{12} \text{ W}^2$.

(c) Now we need to estimate the total variance given by:

$$\begin{aligned} V(P(V_1) + P(V_2)) &= \mathbb{E}[(P(V_1) + P(V_2))^2] - (\mathbb{E}[P(V_1) + P(V_2)])^2, \\ &= V(P(V_1)) + V(P(V_2)) + 2C(P(V_1), P(V_2)). \end{aligned}$$

Since $V(P(V_1)) = V(P(V_2))$, we can then generate directly samples from power function and compute the variance using *var* provided in Matlab.

Since the covariance $2C(P(V_1), P(V_2))$ is already computed in 3(b), we get our estimation of total variance is $3.7369 \times 10^{13}W^2$ and standard deviation of 6.1130×10^6W .

(d) We need to estimate the probability:

$$\mathbb{P}(P(V_1) + P(V_2) > 9.5) \quad \text{and} \quad \mathbb{P}(P(V_1) + P(V_2) < 9.5)$$

Again, we first compute the weighted function $\phi(x, y) \cdot f_{XY}(x, y)$, where ϕ is the target density and try to find an instrumental density function g such that decays similar target function. For both cases, we choose $\mu_g = 12$ and $\sigma_g = 20$ base on observation in 3(b). By drawing 10000 samples from both g function we get the following estimation:

$$\mathbb{P}(P1 + P2 > 9.5e6) : 0.3711 \quad \text{with} \quad 95\%CI : [0.3629, 0.3792]$$

and

$$\mathbb{P}(P1 + P2 < 9.5e6) : 0.6046 \quad \text{with} \quad 95\%CI : [0.5968, 0.6124].$$

Notice that the sum of two probabilities should be 1 and in our case it is 0.9757, one possible reason for this to happen could be the probability of hitting exactly the threshold (9.5e6 W) in numerical evaluation occurs even though the theoretical value should be zero.